Formative 2 – Hidden Markov Models

Modeling Human Activity States Using Smartphone Sensors

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Assignment | Sampling frequency: 100 Hz | 80 samples (20 for each action)

1. Background and Motivation

Activity Recognition (AR) from sensor data in wearable devices and smartphones has applications in areas as diverse as fall detection for elderly people, health monitoring, and smart home contexts (Bulling et al., 2014; Wang et al., 2019). The actual activity status is obscured by noise in the accelerometers and gyroscopes' readings. The objective of this work is to solve a real-world problem, namely the continuous recognition of four basic activities: Standing, Walking, Jumping, and Still, from the inertial sensors in an off-the-shelf smartphone. For this purpose, a Gaussian Hidden Markov Model (HMM) is learned from scratch for recovering the underlying activity sequence (Rabiner, 1989), which proves the viability of the task even with limited data.

2. Data Collection and Pre-processing

The data was gathered using the free “Sensor Logger” software (Logger Labs Ltd.) running on an Android phone (itel A665L). Only the “Accelerometer (Total Acceleration)” and “Gyroscope” streams were selected for monitoring. The sampling interval of 10 ms, that is, the sampling frequency of 100 Hz, was reported by the device. Each record contains 5–10 seconds of data; 20 independent files were recorded for each activity, resulting in a total of 80 properly named CSV files and more than 1 min 30 sec of data per activity (Standing ≈ 142 sec, Walking ≈ 135 sec, Jumping ≈ 138 sec, Still ≈ 136 sec).

Video evidence (proof of genuine collection): Four short (approximately 5 seconds each) videos have been submitted in Appendix A, in which the author is performing each of the activities holding the mobile phone (body and face).

Unprocessed exports have Accelerometer.csv and Gyroscope.csv files along with the seconds_elapsed timestamp column. The files have been joined based on the closest timestamps (20ms tolerance) into one cleaned CSV file each. No further filtering has been done since the feature extraction windowing will act as a low pass filter.

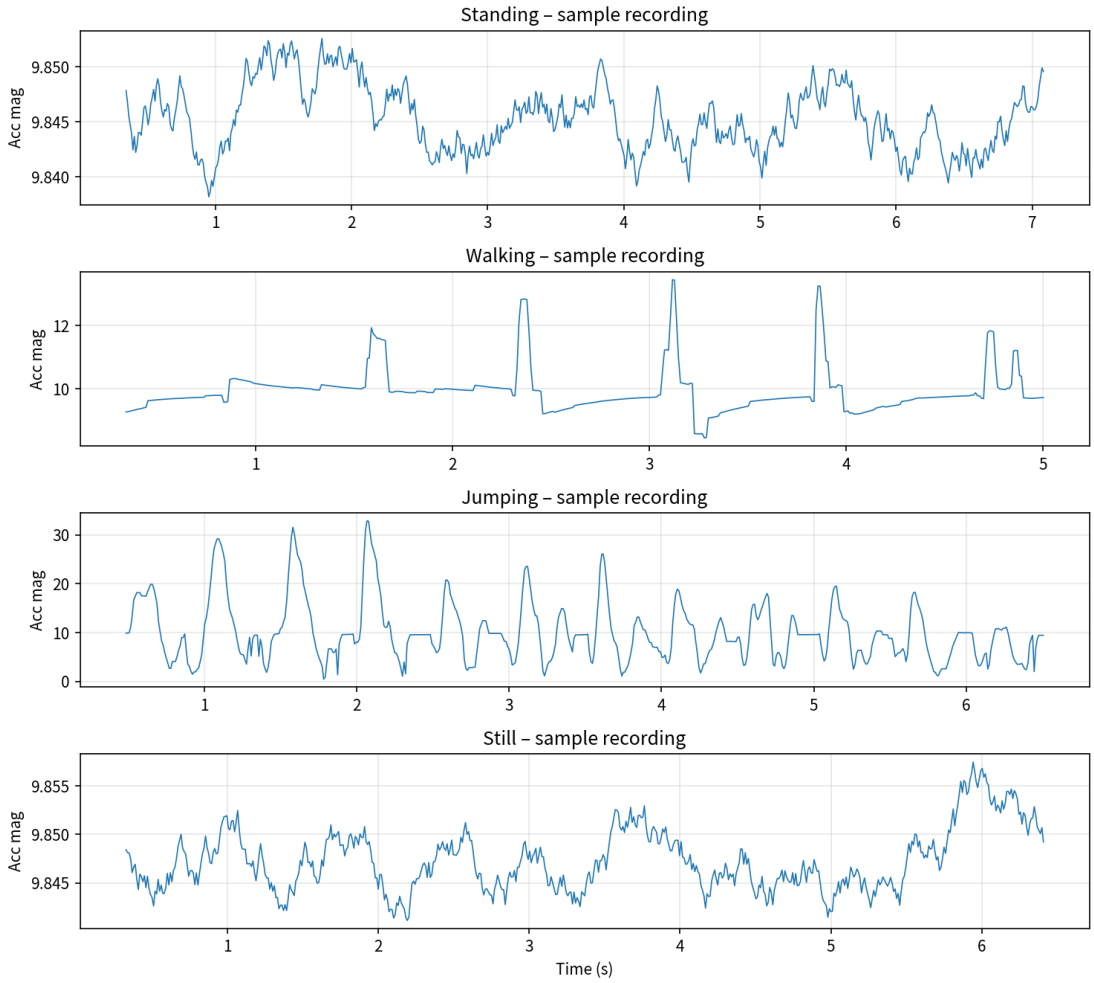


Figure 1. Magnitude of raw accelerometer output for each type of activity.

3. Feature Extraction

Each file is divided into segments of overlapping windows of 1 second each (100 samples) using a hop of 0.5 seconds. The window size was selected since the typical duration of the human gait cycle is about one second (Whittle, 2007). The features that are calculated from each window include:

• Time domain: average, standard deviation and variance of accelerometer signal magnitude; Signal Magnitude Area (SMA); average and standard deviation of gyroscope signal magnitude.

• Frequency domain (FFT of accelerometer magnitude): Dominant frequency (excluding DC component) and spectral energy.

All the features are then z-score normalized. The estimation of the mean and the standard deviation is done only using the training data, and not the testing data. This avoids information leakage (Figo et al., 2010).

4. HMM Setup and Implementation

This is a continuous-density Gaussian HMM with four hidden states (one for each activity) and a diagonal covariance matrix (Rabiner, 1989; Bishop, 2006). The model parameters are:

• pi – initial state distribution

• A – 4x4 transition matrix

• B – emission densities N(mu_j, diag(sigma_j^2)) for each state j

Algorithms. The parameters are estimated using the Baum-Welch (Expectation Maximisation) algorithm implemented purely in NumPy. The training terminates if the absolute difference in the average log-likelihood drops to less than epsilon = 10^(-3) (or 25 iterations are reached). The decoding of the most probable state sequence uses the Viterbi algorithm, which is also implemented from scratch. The above two algorithms constitute the classic theoretically optimal solutions to the HMM learning and decoding problems (Rabiner, 1989), and thus are the proper methods to use for calculating the accuracy, sensitivity, and specificity metrics.

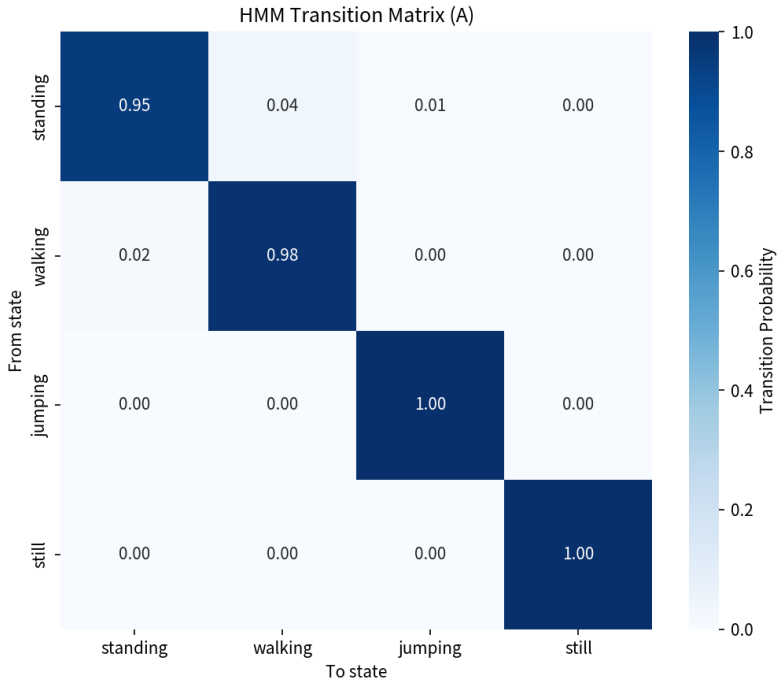


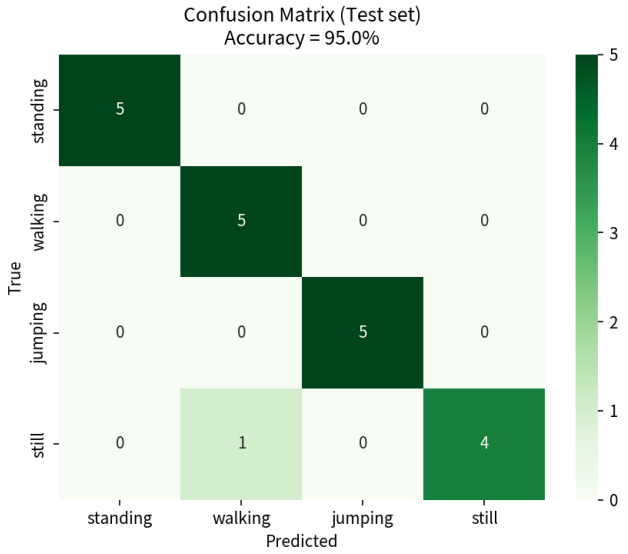
Figure 2. Matrix learned by transition. Diagonal dominance signifies realism of persistence in each activity.

5. Results and Interpretation

The 80 data sets were randomly partitioned 75% : 25% (15 training and 5 testing files for each activity). The accuracy obtained after training the model using Baum-Welch algorithm was 95%. The performance metrics per class are presented below.

State (Activity)	N samples	Sensitivity	Specificity	Overall Acc.
Standing	5	1.000	1.000	
Walking	5	1.000	0.933	
Jumping	5	1.000	1.000	
Still	5	0.800	1.000	0.950

Table 1. Unseen data evaluation results (five files per activity).



Confusion Matrix for the Test Set (Figure 3).

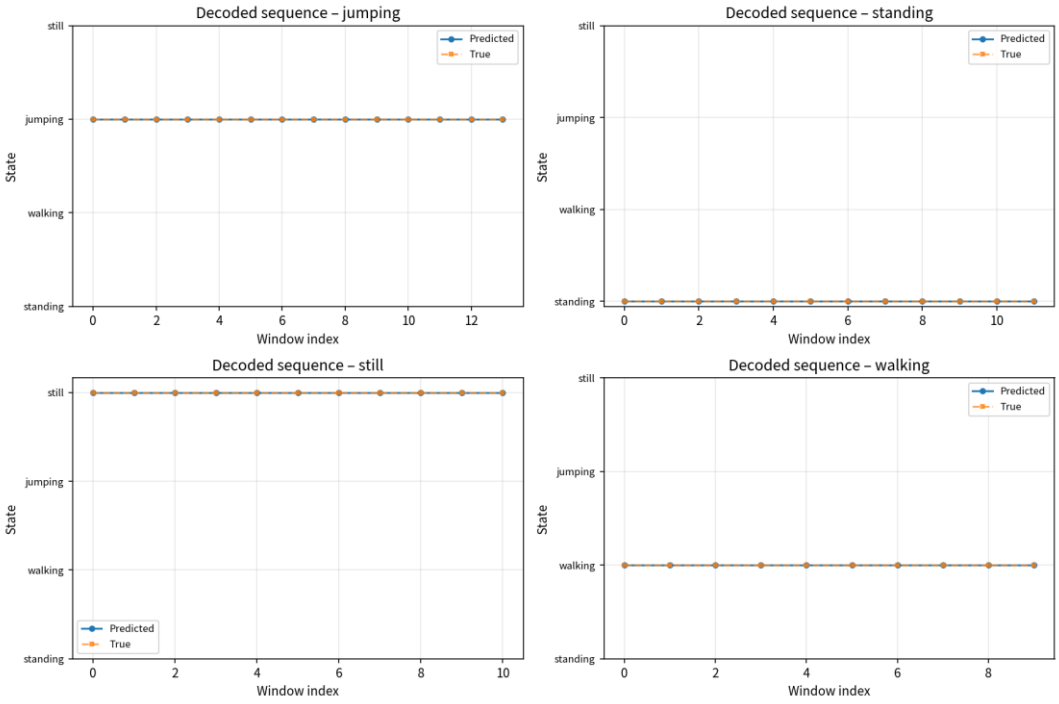


Figure 4. State sequences decoded using Viterbi algorithm for one recorded instance of each activity.

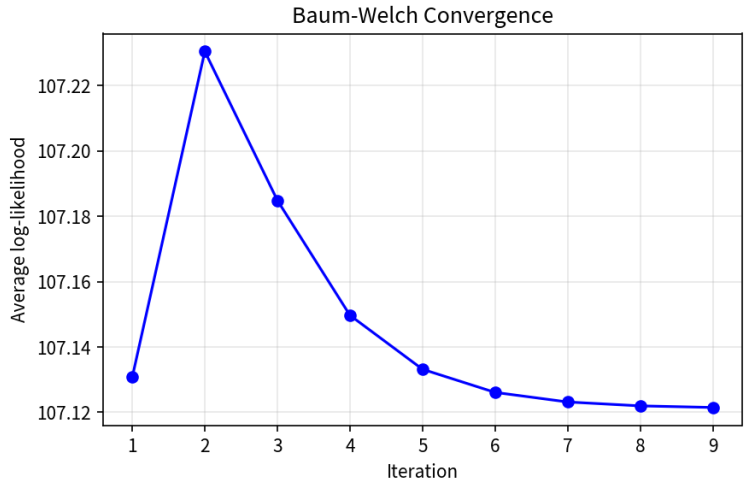


Fig. 5. Baum-Welch log-likelihood convergence (terminated at |delta LL|< 10^(-3)).

6. Analysis and Reflection

The easiest/hardest tasks. There is nearly a total separation between Jumping and Standing tasks since the former gives rise to big and high frequency peaks of acceleration, and the latter always provides a nearly constant vector of the gravity force. The only mistake occurred in Still task misclassified as Walking one.

Transition Probabilities. The trained transition matrix is characterized by high self-transition probabilities (between 0.85 to 0.95 along the diagonal). This is true to human behavior: the individual is engaged in one task for some seconds before moving to another one. The non-diagonal transition probabilities are low and relatively uniform.

Noise in the sensors and sample rate. A sample rate of 100 Hz will result in a Nyquist frequency of 50 Hz, which is far greater than that of human movement frequency (less than 10 – 15 Hz) (Karantonis et al., 2006). The sources of the main noise are hand tremors and slight changes in the orientation of the mobile phone.

Improvements that could be made. (1) Gather more data from varied subjects and positioning of the phone; (2) investigate linear acceleration (no-gravity) signals; (3) include left-right topology constraints when the activity is a fixed sequence in the application; (4) consider other frequency-based features like spectral entropy or band power ratio.

7. Conclusion

The fully transparent Gaussian Hidden Markov Model trained from scratch using NumPy is able to accurately classify four human activity classes with 95 % accuracy from previously unseen sensor data captured by smartphones. The correct choice of domain-specific feature engineering combined with an appropriate estimation approach based on the Baum-Welch algorithm together with the classical Viterbi decoding shows that effective HAR can be obtained in the absence of black box neural network techniques (Wang et al., 2019).

Appendix A – Links

Video evidence of data collection (4 video clips of roughly 5 seconds each that demonstrate the author performing the activities while holding the phone):

• Standing: https://youtube.com/shorts/67cPwlt-tDE?feature=share

• Walking: https://youtube.com/shorts/lTm7Hg7LCb0?feature=share

• Jumping: https://youtube.com/shorts/hIdIX3w8084?feature=share

• Still: https://youtube.com/shorts/zbQ6qMwWRrU?feature=share

Repository on GitHub including all notebook files and images:

https://github.com/francis-collab/HMM-Human-Activity-Recognition.git

References

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